$$\text{mass of the unit cell} = \left\{\begin{matrix}\text{total number of}\\ \text{atoms belongs to}\\ \text{that unit cell}\end{matrix}\right\} \times \left\{\begin{matrix}\text{mass of}\\ \text{one atom}\end{matrix}\right\} \quad \ldots(2)$$

$$\text{mass of one atom} = \frac{\text{molar mass } (\text{gmol}^{-1})}{\text{Avagadro number } (\text{mol}^{-1})}$$

$$m = \frac{M}{N_A} \quad \ldots(3)$$

Substitute (3) in (2)

$$\text{mass of the unit cell} = n \times \frac{M}{N_A} \quad \ldots(4)$$

For a cubic unit cell, all the edge lengths are equal i.e , a=b=c

$$\text{volume of the unit cell} = a \times a \times a = a^3 \quad \ldots(5)$$

$$\therefore \textbf{ Density of the unit cell } \rho = \frac{n\,M}{a^3 N_A} \quad \ldots(6)$$

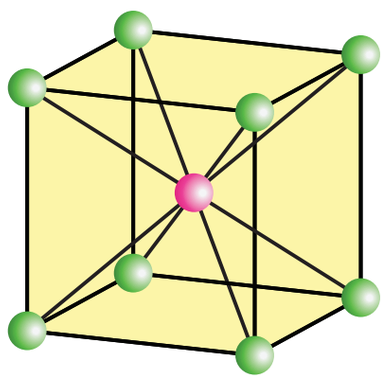
Equation (6) contains four variables namely ρ , n , M and a . If any three variables are known, the fourth one can be calculated.

Example 2

Barium has a body centered cubic unit cell with a length of 508pm along an edge. What is the density of barium in g cm^{-3}?

Solution:

$$\rho = \frac{n\,M}{a^3 N_A}$$



In this case,

n=2 ; M=137.3 $gmol^{-1}$; a = 508pm= 5.08X10^{-8}cm

$$\rho = \frac{2 \text{ atoms} \times 137.3 \text{ g mol}^{-1}}{\left(5.08 \times 10^{-8} \text{ cm}\right)^3 \left(6.023 \times 10^{23} \text{ atoms mol}^{-1}\right)}$$

$$\rho = \frac{2 \times 137.3}{\left(5.08\right)^3 \times 10^{-24} \times 6.023 \times 10^{23}} \text{ g cm}^{-3}$$

$$\rho = 3.5 \text{ g cm}^{-3}$$

XII U6 Solid State - Jerald.indd 185 04-01-2021 13:28:06